SPE-227045-MS

Impact of Hydraulic Fracturing Fluids on Tight Sandstone: Long-Term Behavior and Residue Analysis

Amro Othman, Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; Muhammed Shahzad Kamal and Murtada Saleh Aljawad, Department of Petroleum Engineering, King Fahd University of Petroleum & Minerals, Saudi Arabia / Center for Integrative Petroleum Research, King Fahd University of Petroleum & Minerals, Saudi Arabia; Rajendra Kalgaonkar, EXPECR ARC, Saudi Aramco, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Hydraulic fracturing is a method used to enhance hydrocarbon recovery from tight geological formations by injecting viscous fluids to fracture the rock and deposit proppants. This research investigates the effects of long-term exposure of tight sandstone formations to remaining water-based cross-linked polymer fracturing fluid (FF) under extreme conditions. The objective is to minimize the need for immediate flow-back and cleanup operations. Core flooding experiments are conducted to evaluate the impact of fracturing fluids on core samples and to assess changes in fluid properties over time.

This study used a standard industry fracturing fluid (FF) formulated with low-salinity water (TDS 819 ppm) containing CMHPG polymer, cross-linker, breaker, biocide, pH buffer, surfactant, and gel stabilizer. The FF was broken and filtered before use. Formation water (FW) used for saturation and permeability tests had a TDS of 36,600 ppm. Core samples from Scioto tight sandstone, with permeability ranging from 0.1 to 1.5 millidarcies, had porosity values of 15%-18% measured using gas. Permeability tests were performed at 145°C, 2500 psi confining pressure, and 970 psi back pressure. The cores were saturated with 100 cc of FW, and permeability was measured after FW saturation, FF saturation, and a second FW saturation. Core samples were aged in FF-filled cells at 145°C and 600 psi for extended periods, and permeability was remeasured. Simultaneously, the FF was aged under identical conditions, and different viscosity and characterization tests were applied to the fluid.

The initial permeability measurements revealed a decrease following exposure to the fracturing fluid (FF). After the second saturation with formation water (FW), the permeability slightly improved but remained below the initial value. Long-term aging of the core samples for one and three months resulted in a minimal further reduction in permeability. This pattern was observed both immediately after aging with FF-saturated rock and after subsequent FW flooding. The fracturing fluid and its residue were characterized separately. Rheological properties of the fluid were evaluated before and after breaking and filtering (24 hours) and again after one month of aging. The fluid's color noticeably changed after one month. Residues collected on the filter paper were dried and weighted.

This study highlights the long-term impact of water-based cross-linked polymer fracturing fluids on tight sandstone formations. While permeability decreases initially, minimal further reductions occur after extended aging and repeated flooding. Changes in the fluid's rheological properties and chemical composition over time provide valuable insights for optimizing fracturing fluid formulations, improving well cleanup, and enhancing well performance.

Keywords: Long-term aging, Environmental impact, Fracturing fluids, Residues, Permeability, Core flooding

Introduction

The hydraulic fracturing technique is widely used in large-scale treatments to improve fluid flow in unconventional formations. In these treatments, high-pressure fluids are injected to create fractures in the reservoir formations. The effective fracturing fluids open the fractures and carry the proppants, which keep the fractures open, to allow the easy flow of hydrocarbons. These fluids are designed to minimize pressure loss, reduce fluid leak-off, ensure ease of break and flow back, minimize formation damage, and remain compatible with formation fluids (Das & Rahim, 2014; Samuel et al., 2000). A wide variety of fracturing fluids are tailored to suit specific reservoir conditions. Fracturing fluids are either (a) oil-based fracturing fluids, (b) energized/ foam fracturing fluids, where foaming agents are used with slickwater, cross-linkers, and viscoelastic surfactants to reduce formation damage, (c) Water-based fracturing fluids, including high viscosity systems, which are polymers with cross-linkers and breakers, and slickwater which contains over 90% water. There are other fracturing fluids, including Viscoelastic Surfactants, alcohol and Emulsion fluids, and hybrid fluids that combine slickwater and cross-linked fluids to optimize fluid properties (Barati & Liang, 2014; Fink, 2021; Gaurina-Međimurec et al., 2021).

Fracturing Fluids Additives

These fracturing fluid additives are used to enhance thermal stability, regulate pH, control fluid loss, and minimize formation damage (Harris, 1988). Polymers - the essential components of fracturing fluids- are used to prepare linear gels to condition the formation and initiate the fractures. Once the cross-linker is added, the fluid can be used to initiate and propagate the fracture, and carry the proppant (Fink, 2021; Wilson, 2017). The cross-linkers, such as borate salts, enhance fluid elasticity and viscosity instantly without increasing polymer loading, whereas delayed cross-linkers such as titanium and zirconium oxychloride improve proppant transport (Almubarak et al., 2019; Harris & van Batenburg, 1999; Langford et al., 2013; Powell et al., 1998). Breakers like bromates, peroxy disulfate, and enzymes reduce viscosity after fracture closure to release proppant; however, a controlled breakage is required to reduce the risk of premature screen-out and minimize damage (Almubarak et al., 2019; Denney, 2012; Patil et al., 2013; Sangaru et al., 2017). pH buffers are used to optimize polymer hydration and cross-linker efficiency. Bactericides, which are used to prevent bacterial growth that can release H_2S and fluid degradation (Xue & Voordouw, 2015). The other additives are scale inhibitors, clay stabilizers, surfactants, and acids.

Carboxymethyl Hydroxypropyl Gua-based Fracturing Fluid

The first step in preparing fracturing fluids is to hydrate polymers. In this process, the water molecules penetrate and swell the polymer chains, increasing viscosity. The efficiency of hydration depends on pH, temperature, polymer concentration, and water salinity. It is most effective at the lower pH, close to 5, while extreme pH levels can reduce the hydration efficiency due to polymer degradation or ion interference (Harris et al., 1989). CMHPG is a modified form of guar gum with improved stability and minimal residue buildup. In this modification, carboxymethyl groups are introduced into the guar structure. The guar is treated with propylene oxide and monochloroacetic acid in sodium hydroxide. Controlling the concentration of sodium hydroxide is crucial for the proper modification of the introduced functional groups (Zhang et al., 2005). After hydration, other additives are added to the fracturing fluid.

CMHPG forms strong, stable bonds when cross-linked with zirconium-based cross-linkers; these cross-linked polymers are suitable for high-temperature applications. Their performance deteriorates above 120 °C due to thermal and shear degradation; however, with the addition of synthetic polymers and high-temperature stabilizers like sodium thiosulfate, the thermal stability can be improved up to 177 °C (Almubarak et al., 2021a). These cross-linkers function effectively across a broad pH range (2–12) and temperatures below 38°C to over 204°C, maintaining viscosity in acidic and saline environments (Harris, 1988; Zhou et al., 2021). These systems can exhibit delayed activation, as their effectiveness depends on specific fluid conditions such as temperature, pH, and salinity. (Almubarak et al., 2021b; Fink, 2021). The borate cross-linkers are initially insoluble minerals like borax, which dissolve in water with heat or mixing. This releases borate ions gradually, ensuring cross-linking happens at the right time, to prevent early thickening. Borate ions bond with hydroxyl groups in guar-based polymers, forming reversible bonds. In CMHPG, carboxymethyl groups improve ionic interactions, boosting cross-linking and thermal stability (Bishop et al., 2004). However, their high pH may cause downhole issues such as clay swelling and scale formation, making pH buffering within the range of 8-10 essential for stability. At standard conditions, borate cross-linkers are instant cross-linkers, while at elevated temperatures, they function as delayed cross-linkers because their ion release depends on temperature (de Kruijf et al., 1993; Fink, 2021; Zhou et al., 2021).

Breaking cross-linked gels after placing the proppant allows efficient fluid flowback and minimizes formation damage. Breakers are materials that absorb cross-linkers, leaving minimal residual cross-linkers in the fluid. Breaking the fluid at the correct time improves the fracture conductivity; however, improper breaking can affect proppant placement and lead to premature screen-out. Breakers like bromates, peroxy disulfate, and enzymes reduce viscosity after fracture closure to release proppant (Almubarak et al., 2019; Denney, 2012; Patil et al., 2013; Sangaru et al., 2017). Breakers are categorized into oxidative, enzymatic, delayed, and hybrid systems, that is, both oxidative and enzymatic, each with distinct chemistry and activation mechanisms. The breakers can be introduced to the fracturing fluid in two ways: internal and external breaker addition.

Formation Damage from Polymers

Guar gum and its derivatives, including CMHPG, often leave residues that damage the induced fracture and proppant pack (Brannon et al., 1995). After the fluid viscosity breaks, the fluid forms insoluble mannose helix structures and contains insoluble ash. The proppants react to formation water and fracturing fluid, are crushed due to mechanical stress, and precipitates occur between the proppants and formation fluids. The stress-induced proppant embedment into the fracture face leads to compaction and pore blockage caused by the migration of fine particles. (Karazincir et al., 2019). The fluid residues can affect the wettability of proppant surfaces and nearby rock, which negatively impacts the hydrocarbon flow in the formation (Dong et al., 2019). Polymer residue can form a filter cake that reduces flowback if not fully removed. Gel breaking is often incomplete in low-temperature or high-salinity environments, requiring tailored oxidative or enzyme breakers. Polymers may also trigger clay swelling and fines migration in water-sensitive zones unless treated with stabilizers like potassium chloride. CMHPG provides improved hydration, cleaner breaking, and higher resistance to bacterial growth, leaving minimal residue. Damage control measures include low-residue polymers, effective breakers, and biocides. Advanced fluid designs further enhance performance under thermal and saline stress (Chen et al., 2022; Da et al., 2024; Khan et al., 2021; Lijun et al., 2019).

After the completion of the fracturing treatment, the waiting time for the connection of the well to the gas oil separation plant can take a long time, necessitating flaring activities. As the flow back of the injected fracturing fluids with the associated gases (such as H_2S, methane, and CO_2) and water. However, if the fracturing fluid containing polymers is left in the reservoir for extended periods can cause different sort of damage, which includes reduction in permeability due to adsorption and mechanical precipitation, scaling, and other damages. If the operator can leave the fluid without flowing back, it can eliminate the need for flaring. This study evaluates the long-term effects of water-based cross-linked polymer fracturing fluids on tight sandstone permeability under extreme conditions.

Methodology

Materials

The materials used in this study include the core samples, fracturing fluid components, and formation water. Scioto tight sandstone cores were selected due to their low permeability (0.1–1.5 mD) and consistent porosity (17–20%), making them representative of tight formations. The core sample's dimensions are 1.5-inch diameter × 3-inch length. A cross-linked fracturing fluid was formulated using Low-salinity water from a local well with TDS of 819 ppm, carboxymethyl hydroxypropyl guar (CMHPG) as the base polymer. Two types of buffers are used to reduce pH for better polymer hydration, and the other to raise the pH to activate the instant cross-linker. Two types of cross-linkers, instant and delayed cross-linkers, are based on borate salts and zirconium metals. To break the fluid under harsh conditions, a breaker, a breaker activator, and an incapsulated breaker are used. Biocides, a gel stabilizer, a clay stabilizer, and a surfactant are also used to prepare the fluid. The formation water used is a synthetic formation brine shown in Table 1, with a TDS of 36,600 ppm was prepared and used for permeability testing.

Table 1—Formation water brine

Compound	Concentration (g/L)
$NaHCO_3$	1.03
$CaCl_2$	1.182
NaCl	33.76
$MgSO_4$	0.033
$MgCl_2 \cdot 6H_2O$	0.675
$BaCl_2 \cdot 2H_2O$	0.05
$SrCl_2$	0.078
KCl	0.144
Total	36.952

Equipment

Anton Paar MCR-302 rheometer was used to evaluate the rheology of the fracturing fluid before the system breaks under high temperature. The Vinci core flooding system, with the setup shown in Fig. 2, was used for all the core flow experiments. The high-temperature core flood system consists of Hastelloy parts. The system has two pumps, and the confining and injection pressure can reach up to 10000 psi. The system can handle temperatures up to 150 °C. For the recording of pressure, temperature, and flow data, the system can record the readings every 30 seconds. One of the accumulators is attached outside the system and connected to the core holder to mimic the fluids pumped from the surface at standard conditions to the reservoir.

Figure 1—Vinci core flooding system

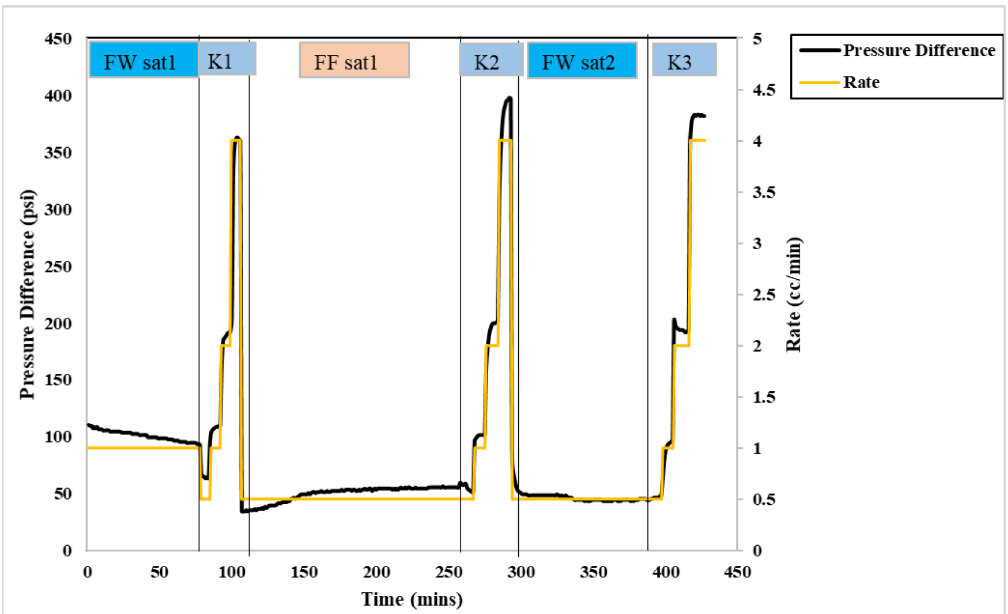

Figure 2—Pressure profile during core flooding stages, Sample 1 (pre-1-month aging)

Procedures

The experimental workflow involved core preparation, fluid preparation, saturation and permeability testing, long-term fluid aging, and the regained permeability evaluation after aging. All core samples were cut with dimensions of 1.5-inch diameter × 3-inch length. They were dried at 80°C, and porosity was determined using helium gas expansion; then, an average pore volume of 16.5 cc was calculated. The FF was prepared and broken inside the oven using a Hastelloy accumulator. Residues collected during filtration were dried and weighed. The filtered fluid was loaded into an accumulator outside the oven. For one core sample that was tested under standard conditions (STD), the core sample was initially saturated with 100 cc of FW at a rate of 1 cc/min, and the permeability was evaluated with FW. Following this, the core sample was saturated with 100 cc of FF from outside the oven at 1 cc/min. Then, the permeability was directly evaluated with FW for the second time. Finally, the core sample was saturated with 100 cc of FW for the second time, and the permeability was evaluated for the third time. The permeability was tested under 970 psi back pressure and 2500 psi confining pressure. For the other core samples, the saturation and permeability evaluation were performed at a temperature of 145°C. The core samples, except the first one, were aged for

1,3, and 6 months inside an accumulator filled with FF under similar conditions of 145°C, and 600 psi for the specified period. Similarly, the fluid was aged for 2 months, and the color, appearance, and viscosity were compared before and after breaking and aging.

Experiment Design

One batch of the fracturing fluid is prepared, and its rheology is evaluated. The fluid viscosity is then broken at 145 ^{o}C, filtered, and then aged for 2 months under the same conditions separately. Other batches are prepared, broken, filtered then used for the core flooding experiments. The core samples are taken from the same block of Scioto sandstone, dried, weighed, and measured for length. The following table 1 shows the core flooding experiments matrix.

Table 2—Core flooding experiment set up

No of Cores	Test Temperature (^{o}C)	Test back pressure (psi)	Duration (Months)
1	25	1000	6
3	145	970	1.3 and 6

Results and Discussions

One-Month Sample at HPHT

The first core flooding experiment was conducted at 145 °C, and the core sample was aged at 145 °C and 500 psi for one month, then the core flooding was performed again. At the first round of flooding, the core was subjected to formation water saturation abbreviated in figures by (FW sat 1), followed by permeability measurement (K1). Then, CMHPG fracturing fluid saturation (FF sat), followed by the second permeability measurement (K2), and a final FW saturation (FW sat 2), then the third permeability measurement (K3). The permeability measurements were K1 = 0.57 mD, K2 = 0.49 mD, and K3 = 0.52 mD. After the measurements finish, the core sample is saturated with fracturing fluid and saved in an accumulator and aged for one month. Comparing K3 to K1, a minor permeability reduction was observed upon exposure to the fracturing; 91% of its original permeability was regained. Figure 2 shows the pressure difference changes during the stages of the core flooding experiment. It can be noted that a higher-pressure difference was noted in the FW saturation part, as 1 cc/min was applied compared to 0.5 cc/min for the saturation of the other parts. After aging the system to record post-aging pressure data, however, the live recording of after FW saturation was captured, and the calculated permeability was 0.51 mD, corresponding to a 90% regained permeability. This suggests that the fracturing fluid caused only minor and largely reversible damage to the tight sandstone formation during one month of aging. The similarity between the pre-aging and post-aging regained permeability values highlights the stability of the CMHPG-based system and its low residue impact under the tested reservoir conditions.

Three-Month Sample Prior Aging

The second core flooding experiment was conducted at 145 °C, before and after 3 months of aging. Following FW sat 1, K1 was measured at 0.55 mD; after FF sat, K2 dropped to 0.28 mD; and after FW sat 2, K3 increased to 0.38 mD. The regained permeability before aging was calculated to be 69 % relative to K1. This limited recovery is attributed in part to the quality of the fluid used, which may have contributed low quality of breaking, although it was filtered several times. Figure 3 shows the pressure difference recorded throughout the FW and FF saturation stages, which were pumped at 1 cc/min, and during permeability measurements. Notably, the rise in pressure observed during FF sat, as shown in Figure 3, reflects the impact of fluid quality and potential formation damage. The core sample was not blocked; therefore, the sample was aged, and the test continued.

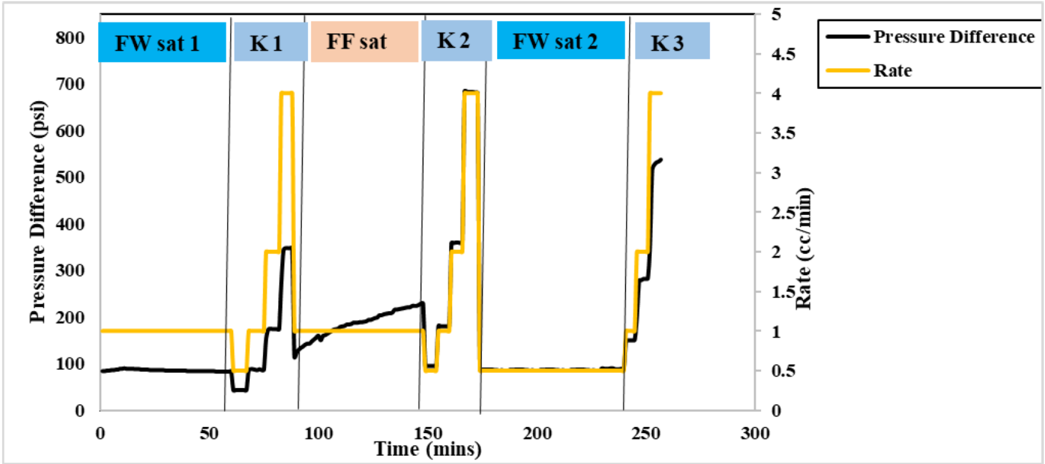

Figure 3—Pressure profile during core flooding stages, Sample 2 (pre-3 months aging)

Three-Month Sample after Aging at HPHT

After aging, the test began with permeability directly after aging (KAA), which was 0.25 mD, indicating a lower value compared to the same measurement before aging. Then FW sat was performed, and the permeability measured after this saturation (KA FW) showed a slight increase, reaching 0.28 mD. The regained permeability, calculated relative to the initial value before aging, was 51% compared to the 69 % recovery recorded before aging. It can be noticed that, 18 % reduction in regained permeability. Figure 3a illustrates the pressure difference across the core during the various flooding stages. IT was noticed that a stable and slight gradual decrease in pressure difference during the FW sat. However, the pressure difference value was high, 200 psi, compared to 100 psi in the saturation of the other core samples. That rose is attributed to the fluid quality discussed earlier.

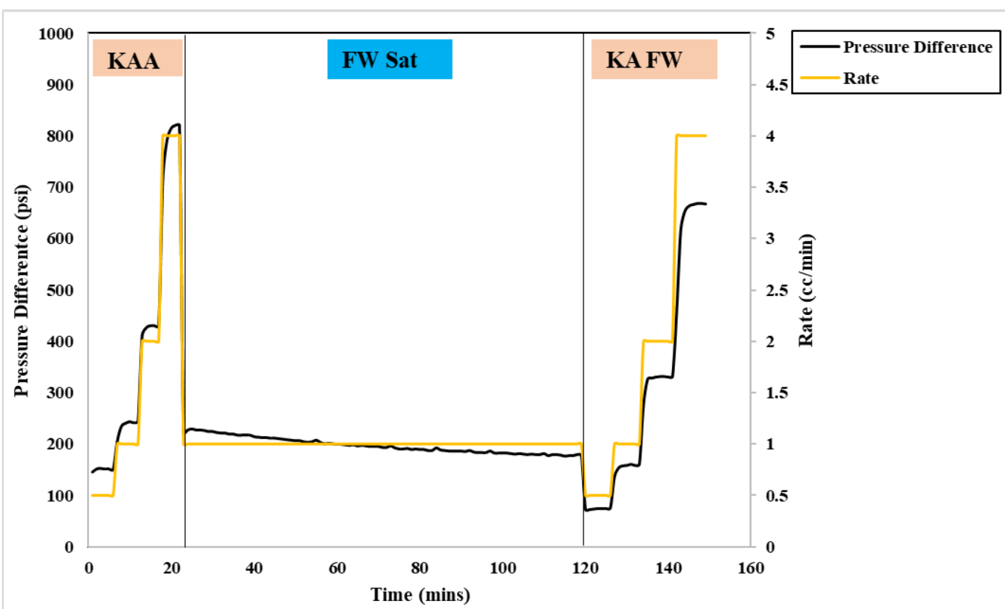

Figure 3a—Pressure profile during core flooding stages, Sample 2 (after 3 months aging)

Six-Month Sample Prior Aging

The fourth core flooding experiment was conducted at 145 °C, before and after 6 months of aging. Following FW sat 1, K1 was measured at 0.58 mD; after FF sat, K2 decreased to 0.41 mD; and after FW sat 2, K3 increased to 0.46 mD. The regained permeability before aging was calculated to be 79 % relative to K1.

This recovery indicates that while the fracturing fluid caused an initial reduction in permeability—likely due to polymer invasion or filter cake formation—a significant portion of the damage was reversible. Figure 4 shows the pressure difference recorded during FW and FF saturation stages, which were pumped at 1 cc/min, along with the associated permeability measurements. A pressure rise was observed during FF sat and permeability measurement, highlighting the effect of the initial damage to the FF on the formation. The damage was significantly lower than what was observed in core 2, as the pressure difference was 100+ psi compared to 200+ psi. This can also be noted from the regained permeability values.

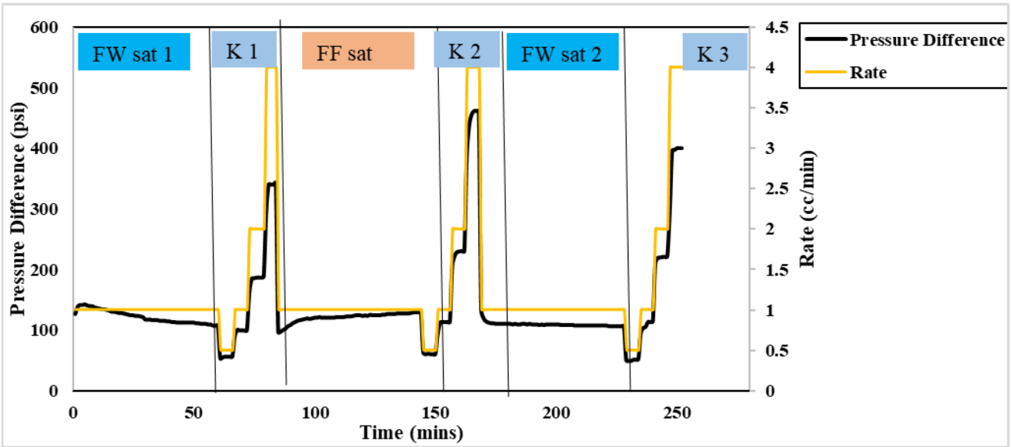

Figure 4—Pressure profile during core flooding stages, Sample 3 (pre-6 months aging)

Six-Month Sample after Aging at HPHT

After aging the same core sample for 6 months in CMHPG-based fracturing fluid at 145 °C and 600 psi, the test began with permeability measured KAA, which was 0.29 mD, showing a reduction compared to the pre-aging value (0.41mD). Then FW sat was performed, and the permeability measured after this saturation (KA FW) showed a slight increase, reaching 0.34 mD. The regained permeability, calculated relative to the initial value before aging, was 59 %, indicating a 20 % reduction from the 79 % recovery achieved before aging. Figure 5 illustrates the pressure difference across the core during the flooding stages.

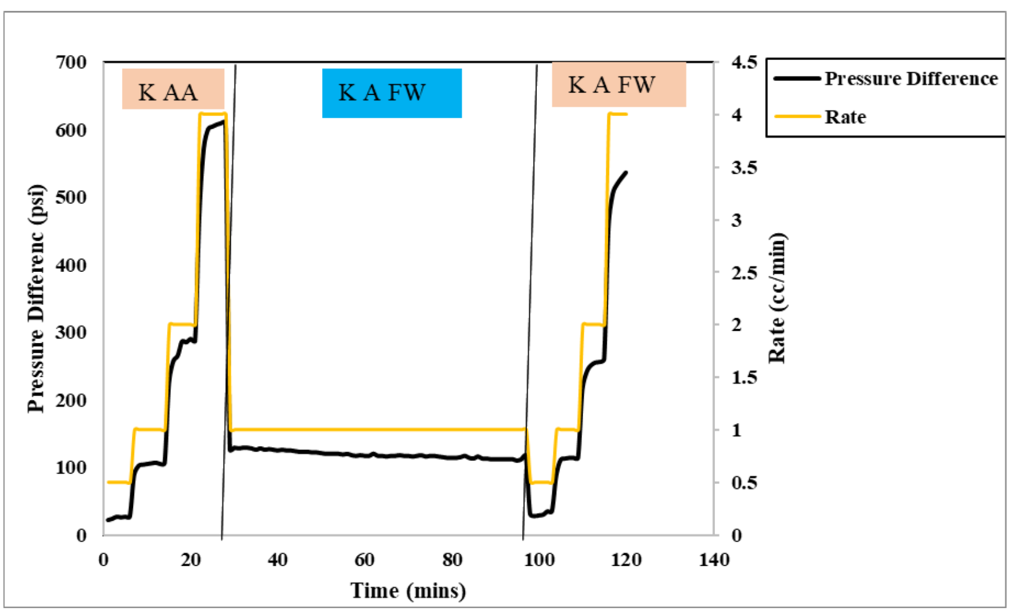

Figure 5—Pressure profile during core flooding stages, Sample 3 (after 6 months aging)

Six-Month Sample Prior Aging at STD

The fifth core flooding experiment was conducted under standard conditions to evaluate the impact of CMHPG-based fracturing fluid on permeability before any aging. The test followed a three-stage procedure, beginning with FW sat 1, after which the permeability was measured: K1 = 0.90 mD. This was followed by FF sat, and the second permeability measurement was taken: K2 = 0.61 mD. Finally, FW sat 2 was carried out, and the third permeability was measured: K3 = 0.66 mD. Comparing K3 to K1, the regained permeability was calculated to be 73 %, indicating that the fracturing fluid caused a noticeable reduction in flow capacity, likely due to polymer residue or filter cake formation. However, the partial recovery in K3 reflects that the damage was, to some extent, reversible under these conditions. Although the test was conducted at standard conditions without thermal or pressure aging, the permeability drop between K1 and K2 highlights that polymer-based fluids can still impair formation permeability in the short term. The improvement observed after FW sat 2 suggests that fluid cleanup using formation water was moderately effective, but not complete. These findings underline the importance of optimizing polymer breaking and cleanup strategies even at standard conditions, as residual damage may persist if not properly addressed. The permeability data and test durations are summarized in the following table 3.

Table 3—Permeability and Regained Values Before and After Aging

Condition	K before FF (md)	K directly after FF (md)	K after FW (md)	Regained K after FF (%)	Regained K after aging (%)
Initial	0.57	0.49	0.52	91	
After 1 month 145 ºC		-	0.51		90
Initial	0.55	0.28	0.38	69	
After 3 months, 145 ºC		0.25	0.28		51
Initial	0.58	0.41	0.46	79	
After 6 months, 145 ºC		0.29	0.34		59
Initial	0.9	0.61	0.66	73	
After 6 months, STD		0.493	0.5		56

The results show that aging CMHPG-based fracturing fluids in the formation reduces the regained permeability over time. One month duration did not affect the formation permeability; however, after 3 months, the regained permeability drops by 17-20% depending on the breaking efficiency of the FF. Nevertheless, the most significant permeability loss occurs immediately after the fracturing fluid is introduced, even before aging. In some cases, permeability drops by over 30% directly after fluid saturation, highlighting that the fracturing fluid itself is the primary source of initial formation damage, even when flowback is performed right away. While aging worsens the damage, these results emphasize that minimizing the initial impact of the fracturing fluid is just as important as preventing long-term exposure, especially in scenarios where flowback is delayed. Designing low-damage, low-residue fluids is therefore critical to preserving formation integrity.

Viscous, Broken, and Aged Fluid

The sequence shown in Figure 6 demonstrates the fluid's thermal breakdown and color change over time. This indicates the change in properties with time. The breakdown and aging are conducted at 145 °C. Image (a) presents the initial viscous CMHPG-based fracturing fluid (FF). In (b), the fluid is partially broken after 3 hours, showing phase separation and precipitate formation. Image (c) shows the fully broken fluid after 24 hours; at this stage, the fluid is filtered, and the residue is weighed. The weighted residue was 2.2 g from a total fluid of 1393g. The Image (d) displays the fluid after filtering and aging for two months, showing a dark brown color and sediment, reflecting advanced chemical changes. It seems the fluid gets darker and burns with time, which affects the rheology and subsequently the formation permeability. The residue percentage was slightly less than 0.2% which is considered very few compared to other guar derivatives. The fluid was filtered, and yet there was sort of damage in the core samples; therefore, it can be suggested that there are other sorts of reasons that cause the damage to the rock samples.

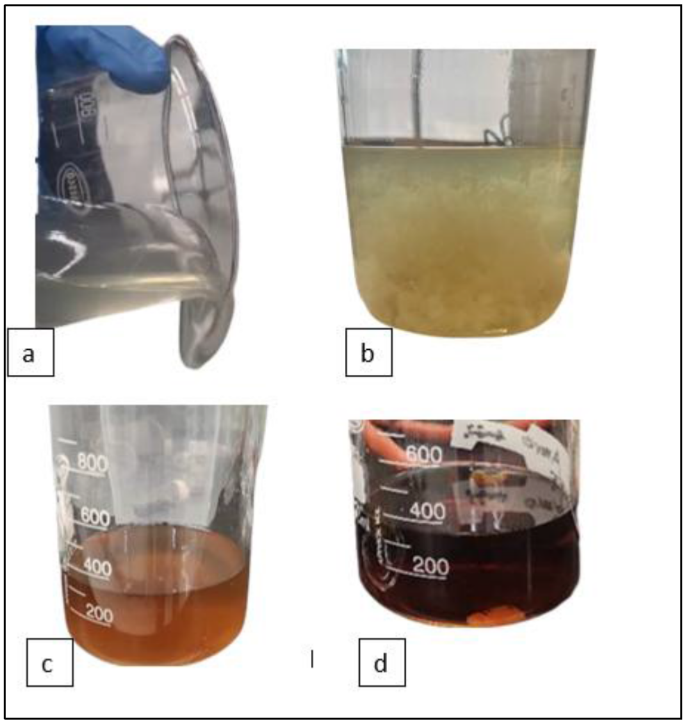

Figure 6—CMHPG-based fluid—(a) viscous; (b) broken after 3 h at 145 °C; (c) broken after 24 h at 145 °C; (d) filtered and aged 2 months

Conclusions

This study aims to evaluate the long-term impact of CMHPG-based fracturing fluids on the permeability of tight sandstone formations under high-temperature conditions. The goal is to assess potential formation damage by performing core flooding on the core samples under reservoir conditions. The FF were broken, filtered, and used for prior and after aging the core samples. The following points show the main conclusions of this study.

1. CMHPG-based fracturing fluids cause immediate permeability damage before aging starts.
2. At short aging durations, full recovery of permeability can be achieved, with an aging duration increase the regained permeability decreases to reach a 20% reduction after 6 months.
3. Aging the core samples for 6 months at STD also damages the formation with a slightly smaller percentage.
4. Visual observation of FF from viscous to 2-month aging revealed that the fluid thermally degrades with time, and the color gets darker, which damages the formation further.
5. The damage to the formation is not only caused by the fluid mechanical precipitation, but there are other chemical causes.

Acknowledgments

The authors would like to acknowledge the College of Petroleum and Geosciences (CPG), King Fahd University of Petroleum & Minerals, for supporting this research under CPG21106. Many thanks to Halliburton for supplying the chemicals.

References

Almubarak, T., AlKhaldi, M., Ng, J. H., & Nasr-El-Din, H. A. (2019). Design and application of high-temperature raw-seawater-based fracturing fluids. *SPE Journal*, **24**(4), 1929–1946. https://doi.org/10.2118/195597-PA

Almubarak, T., Li, L., Ng, J. H., Nasr-El-Din, H., & AlKhaldi, M. (2021a). New insights into hydraulic fracturing fluids used for high-temperature wells. *Petroleum*, **7**(1), 70–79. https://doi.org/10.1016/j.petlm.2020.05.002

Almubarak, T., Li, L., Ng, J. H., Nasr-El-Din, H., & AlKhaldi, M. (2021b). New insights into hydraulic fracturing fluids used for high-temperature wells. *Petroleum*, **7**(1), 70–79. https://doi.org/10.1016/j.petlm.2020.05.002

Barati, R., & Liang, J. T. (2014). A review of fracturing fluid systems used for hydraulic fracturing of oil and gas wells. *Journal of Applied Polymer Science*, **131**(16), 1–11. https://doi.org/10.1002/app.40735

Bishop, M., Shahid, N., Yang, J., & Barron, A. R. (2004). Determination of the mode and efficacy of the cross-linking of guar by borate using MAS11B NMR of borate cross-linked guar in combination with solution11B NMR of model systems. *Dalton Transactions*, **17**, 2621–2634. https://doi.org/10.1039/b406952h

Brannon, H. D., Tjon-Joe-Pin, R. M., Services, B. J., & Usa, C. (1995). Application of Polymeric Damage Removal Treatment Results in Multi-Fold Well Productivity Improvement: A Case Study. *SPE Middle East Oil Show*. http://onepetro.org/SPEMEOS/proceedings-pdf/95MEOS/95MEOS/SPE-29822-MS/4009973/spe-29822-ms.pdf/1

Chen, M., Yan, M., Kang, Y., Fang, S., Liu, H., Wang, W., Shen, J., & Chen, Z. (2022). Shale Formation Damage during Fracturing Fluid Imbibition and Flowback Process Considering Adsorbed Methane. *Energies*, **15**(23). https://doi.org/10.3390/en15239176

Da, Q., Yao, C., Zhang, X., Li, L., & Lei, G. (2024). Reservoir Damage Induced by Water-Based Fracturing Fluids in Tight Reservoirs: A Review of Formation Mechanisms and Treatment Methods. In *Energy and Fuels*. American Chemical Society. https://doi.org/10.1021/acs.energyfuels.3c04963

Dong, K., Wang, M., & Zhang, C. (2019). Effect of wettability of ceramic proppant surface in guar gum solution on the oil flow efficiency in fractures. *Petroleum*, **5**(4), 388–396. https://doi.org/10.1016/j.petlm.2018.12.009

Das, P., & Rahim, Z. (2014). Evaluate fracturing fluid performance for hydraulic stimulation in Pre-Khuff sandstone reservoirs of Ghawar gas field. Society of Petroleum Engineers - SPE Saudi Arabia Section Technical Symposium and Exhibition, **2002**. https://doi.org/10.2118/172217-ms

de Kruijf, A. S., Roodhart, L. P., & Davies, D. R. (1993). Relation between chemistry and flow mechanics of borate-crosslinked fracturing fluids. *SPE Production and Facilities*, **8**(3), 165–169. https://doi.org/10.2118/25206-pa

Denney, Dennis. J. (2012). Fracturing-Fluid Effects on Shale and proppant embedment. *Ournal of Petroleum Technology* **64.03** (2012): 59–61., March.

Fink, J. (2021). Petroleum Engineer's Guide to Oil Field Chemicals and Fluids. In *Gulf Professional Publishing* (3rd ed.). Gulf Professional Publishing.

Gaurina-Međimurec, N., Brkić, V., Topolovec, M., & Mijić, P. (2021). Fracturing fluids and their application in the Republic of Croatia. In *Applied Sciences* (Vol. **11**, Issue 6). MDPI. https://doi.org/https://doi.org/10.3390/app11062807

Harris, P. C. (1988). Distinguished Fracturing-Fluid Additives. October, 1979–1981.

Harris, P. C., Harms, W. M., Norman, L., & Services, H. (1989). Study of Continuously Mixed Crosslinked Fracturing Fluids With a Recirculating Flow-Loop Viscometer. *SPE Production Engineering*. http://onepetro.org/PO/article-pdf/4/04/430/2633041/spe-17044-pa.pdf/1

Harris, P. C., & van Batenburg, D. (1999). Comparison of freshwater- and seawater-based borate-crosslinked fracturing fluids. Proceedings - SPE International Symposium on Oilfield Chemistry, 671–675. https://doi.org/10.2523/50777-ms

Karazincir, O., Li, Y., Zaki, K., Williams, W., Wu, R., Tan, Y., Rijken, P., & Allan, C. ; (2019). Measurement of Reduced Permeability at Fracture Face Due to Proppant Embedment and Depletion-Part II. SPE Annual Technical Conference and Exhibition, Calgary, Alberta, Canada, September 2019., **2**, 196204. https://doi.org/https://doi.org/10.2118/196204-MS

Khan, H. J., Spielman-Sun, E., Jew, A. D., Bargar, J., Kovscek, A., & Druhan, J. L. (2021). A Critical Review of the Physicochemical Impacts of Water Chemistry on Shale in Hydraulic Fracturing Systems. *Environmental Science and Technology*, **55**(3), 1377–1394. https://doi.org/10.1021/acs.est.0c04901

Langford, M., Holland, B., Green, C. A., Bocaneala, B., & Norris, M. (2013). Offshore horizontal well fracturing: Operational optimisation in the southern North Sea. Society of Petroleum Engineers - SPE Offshore Europe Conference and Exhibition, OE 2013, **2**, 104–113. https://doi.org/10.2118/166550-ms

Lijun, Y., Benbin, X., Jian, Y., Yili, K., Huifen, H., Liang, W., & Bin, Y. (2019). Mechanism of fracture damage induced by fracturing fluid flowback in shale gas reservoirs *,**-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/). https://doi.org/10.3787/j.issn

Patil, P., Muthusamy, R., & Pandya, N. (2013). Novel controlled-release breakers for high-temperature fracturing. *Society of Petroleum Engineers - North Africa Technical Conference and Exhibition 2013, NATC 2013*, **1**(April), 599–604. https://doi.org/10.2118/164656-ms

Powell, R. J., McCabe, M. A., Slabaugh, B. F., Terracina, J. M., & McPike, T. (1998). Gulf of Mexico frac-and-pack treatments using a new fracturing fluid system. *World Oil*, **219**(11), 69–73. https://doi.org/10.2523/39897-ms

Sangaru, S. S., Yadav, P., Huang, T., Agrawal, G., & Chang, F. F. (2017). Surface modified nanoparticles as internal breakers for viscoelastic surfactant based fracturing fluids for high temperature operations. Society of Petroleum Engineers - SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition 2017, 2317–2323. https://doi.org/10.2118/188050-ms

Samuel, M., Polson, D., Graham, D., Kordziel, W., Waite, T., Waters, G., Vinod, P. S., Fu, D., & Downey, R. (2000). Viscoelastic surfactant fracturing fluids: Applications in low permeability reservoirs. SPE Rocky Mountain Regional Meeting Proceedings, 1–7. https://doi.org/10.2523/60322-ms

Wilson, A. (2017). A Comparison Between Seawater-Based and Freshwater-Based Fracturing Fluids. *Journal of Petroleum Technology*, **69**(03), 46–47. https://doi.org/10.2118/0317-0046-jpt

Xue, Y., & Voordouw, G. (2015). Control of microbial sulfide production with biocides and nitrate in oil reservoir simulating bioreactors. *Frontiers in Microbiology*, **6**(DEC). https://doi.org/10.3389/fmicb.2015.01387

Zhang, L. M., Zhou, J. F., & Hui, P. S. (2005). A comparative study on viscosity behavior of water-soluble chemically modified guar gum derivatives with different functional lateral groups. *Journal of the Science of Food and Agriculture*, **85**(15), 2638–2644. https://doi.org/10.1002/jsfa.2308

Zhou, M., Zhang, J., Zuo, Z., Liao, M., & Peng'ao, P. (2021). Preparation and property evaluation of a temperature-resistant Zr-crosslinked fracturing fluid. *Journal of Industrial and Engineering Chemistry*, **96**, 121–129. https://doi.org/10.1016/j.jiec.2021.01.001